DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF SCIENCE & COMPUTER STUDIES
MATHEMATICS AND STATISTICS DEPARTMENT

COURSE SYLLABUS

COURSE CODE	:S-MATH 200
COURSE TITLE	:Number Theory
COURSE TYPE	:Lecture
COURSE CREDIT	:Three units
LEARNING MODE	:Traditional Blended
PRE/CO-REQUISITES	:None
PROFESSOR	:
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VISION

A leading Catholic University that inspires excellence and drives innovation towards a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

This course gives importance to the development of basic concepts in arithmetic and algebra as well as the application of relevant concepts and techniques to learn in class with the use of technology. It then proceeds to discuss reading and writing proofs to be creative and innovative in applying theorems effectively and efficiently. It includes topics on fundamental properties of integers, methods of proof, divisibility, primes and non- primes, greatest common divisor, least common multiple, Euclidean algorithm, Diophantine equations, theorems on modulo and congruences, the Chinese remainder theorem.

The course will have a special provision on modality, pacing, pedagogy, assessment type, communication and feedback based on national, regional, provincial, city, and institutional guidelines for the safety amidst disaster and pandemic. Hence, this class shall be delivered in an online module-based approach. This course is also compliant with Outcomes-Based Education (OBE) and 21st Century Learning Design (21CLD) frameworks.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO1. value the importance of constructing proof using fundamental properties of integers;

CLO2. apply the four methods of proof, summation and products, binomial expansions;

CLO3. select the appropriate different properties used in proving theorems;

CLO4. prove the different theorem such as GCD, Euclidean algorithm, Diophantine equations, relatively prime integers, and LCM;

CLO5. investigate the validity of the technique and proof of congruence theorem; and

CLO6. demonstrate 21st century skills in all learning activities;

LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Utilize the importance of constructing proof using fundamental properties of integers CLO2. Apply the 4 methods of proof summation and products, binomial expansion	• TLO1. Define the theory operations and properties • TLO2. Explain the basic properties of integers, well-ordering principle. • TLO3. Prove the given statements using the 4 methods of proof.
CLO3. Evaluate the appropriateness of the properties used in proving theorems.	• TLO4. Identify the basic properties of divisibility, division in the set of integers • TLO5. Construct proofs using the theorems of divisibility.
CLO4. State and prove the GCD, Euclidean algorithm, Diophantine equations, relatively prime integers, and LCM.	• TLO6. Construct proofs using the theorems of GCD, relatively prime integers and LCM • TLO7. Apply / prove the theorem/ definition of Euclidean Algorithm GCD , LCM, and relatively prime integers.
CLO6. Demonstrate 21st century skills in all learning activities.	• *Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module Title and Learning Outcome(s)	**Week No.**	**Teaching-Learning Activities / Assessment Strategy/ Online Activities**	**Technology Enabler**	**Onsite**	**Asynchronous**	**Allotted Hours**

Module 1: Course Introduction and Review on Fundamental Properties of Integers TLO1 TLO2 TLO3 TLO4 CLO1 CLO6	Week 1 Aug 19-24	• Gospel Reading, Prayer and Reflection Module 1 consists of the following: • Course Overview • Course requirement details • About the Professor • Discussion Module 2 consists of the following topics: Fundamental properties of integers Properties of well ordering principles • Study uploaded materials on Module 1 for reinforcement learning. • Watch uploaded supplementary videos about fundamental properties of integers. Enabling Assessment (1): Online quiz on properties of integers	Schoolbook MS Teams		1.5	1.5
		• Gospel Reading, Prayer and Reflection • Attend onsite class for discussion on properties of integers (application)		1.5		1.5
Module 2: Methods of Proof TLO1 TLO2 TLO3 CLO1 CLO6 CLO2	Week 2 Aug 26-31	• Gospel Reading, Prayer and Reflection • Study uploaded materials on Module 2-A: consists of the following topics: 4 Methods of Proof • Watch uploaded supplementary videos about methods of proof	Schoolbook MS Teams		1.5	1.5
		• Gospel Reading, Prayer and Reflection • Attend onsite class for discussion of 4 methods of proof •		1.5		1.5
	Week 3 Sep 2-7	• Gospel Reading, Prayer and Reflection • Module 2 consists of the following topics: Direct Proof Indirect Proof Proof by contradiction Mathematical Induction • Study uploaded materials on Module 2-B Direct Proof, Indirect Proof, Proof by contradiction • Watch uploaded supplementary videos about Direct Proof, Indirect Proof, Proof by contradiction	Schoolbook MS Teams		1.5	1.5

		• Gospel Reading, Prayer and Reflection • Attend onsite class for discussion and participate in onsite seatwork (1) on direct proof, indirect proof, proof by contradiction		1.5		1.5
	Week 4 Sep 9 – 14	• Gospel Reading, Prayer and Reflection Take onsite Enabling Assessment (2): Direct proof, indirect proof and proof by contradiction • Study uploaded materials on Module 2: topics consist of: mathematical induction, binomial expansion, summation & products	Schoolbook MS Teams		1.5	1.5
		• Gospel Reading, Prayer and Reflection • Attend onsite class for discussion on mathematical Induction and binomial expansion		1.5		1.5
Module 3 Divisibility TLO4 TLO5 TLO7 CLO3	Week 5 Sep 16-21	• Gospel Reading, Prayer and Reflection • Module 3 consists of the topics: Divisibility Proving theorem • Study uploaded materials on Module 3: divisibility lesson • Watch uploaded supplementary videos on divisibility	Schoolbook MS Teams		1.5	1.5
		• Gospel Reading, Prayer and Reflection • Attend onsite class for the discussion on divisibility.		1.5		1.5
	Week 6 Sep 23 – 28	• Gospel Reading, Prayer and Reflection • Study uploaded materials on Module 3: Division in the set of integers • Watch uploaded supplementary videos about division in the set of integers	Schoolbook MS Teams		1.5	1.5
		• Gospel Reading, Prayer and Reflection • Take online Enabling Assessment (3): on divisibility, division in the set of integers		1.5		1.5
Module 4	Week 7 Sep 30 – Oct 5	• Gospel Reading, Prayer and Reflection • Study uploaded materials on Module 4: GCD Proving theorems • Watch uploaded supplementary videos about GCD	Schoolbook MS Teams		1.5	1.5

Greatest Common Divisors, Relatively Prime Integers TLO6 TLO7 CLO4		• Gospel Reading, Prayer and Reflection • Attend onsite class for discussion and participate in onsite seatwork(2) in proving GCD theorems		1.5		1.5
	Week 8 Oct 7 – 12	• Gospel Reading, Prayer and Reflection • Study uploaded materials on Module 4: Relatively Prime Integers • Watch uploaded supplementary videos about relatively Prime Integers	Schoolbook MS Teams		1.5	1.5
		• Gospel Reading, Prayer and Reflection • Take online Enabling Assessment(4): Onsite quiz on GCD		1.5		1.5
	Week 9 Oct 14-19	Review topics under Modules 1 – 3			1.5	1.5
		Summative Assessment: 1 Midterm Exam Onsite Paper and Pen Exam on Topics Modules 1 – 3		1.5		1.5
			TOTAL	**12**	**15**	**27.0**

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO4. Prove the different theorems such as GCD, Euclidean algorithm, Diophantine equations, relatively prime integers, and LCM	• TLO8. Define and prove the Euclidean algorithm. • TLO9. Prove and apply the Diophantine equations. • TLO10. Apply the fundamental theorem of arithmetic.
CLO5. Investigate the validity of the technique and proof of the congruence theorems	• TLO11. Define and prove the basic properties/ theorems of congruences. • TLO12. Apply the theorem of linear congruences to find the solutions. • TLO13. Define and apply the formula of Chinese Remainder Theorem.
CLO 6. Demonstrate 21st century skills in all learning activities.	• *Collaboration, Skilled Communication, Knowledge Construction, Self- Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module Title and Learning Outcome(s)	**Week No.**	**Teaching-Learning Activities / Assessment Strategy/ Online Activities**	**Technology Enabler**	**Onsite**	**Asynchronous**	**Allotted Hours**

Module 5 Euclidean Algorithm, LCM TLO8 CLO4 CLO6	Week 10 Oct 21-26	**Synchronous Week** • Gospel Reading, Prayer and Reflection • Module 4 consists of the following topics: Euclidean Algorithm, LCM • Study uploaded materials on Module 4: Euclidean Algorithm, ,LCM • Watch uploaded supplementary videos about Euclidean Algorithm, LCM	Schoolbook MS Teams			3.0
		• Gospel Reading, Prayer and Reflection • Attend onsite class for discussion of Euclidean Algorithm				
	Week 11 Oct 28-Nov 2	**Independent Learning Days** • Gospel Reading, Prayer and Reflection • Watch the video on how to prove theorems of LCM & Diophantine Equations • Solve Online Exercises for practice			3.0	3.0
	Week 12 Nov 4 – 9	• Gospel Reading, Prayer and Reflection • Study uploaded materials on Module 5: Least Common Multiple • Solve Online Exercises for practice Least Common Multiple	Schoolbook MS Teams		3.0	3.0
		• Attend onsite class for the discussion of the Least Common Multiples				
Module 6-Diophantine equations, Congruences, TLO9 CLO3 CLO5 CLO6	Week 13 Nov 11 -16	• Gospel Reading, Prayer and Reflection • Module 5 consists of the topic: Diophantine Equations • Study uploaded materials on Module 6: Proving Diophantine equations • Watch uploaded supplementary videos about Diophantine equations	Schoolbook MS Teams		1.5	1.5
		• Gospel Reading, Prayer and Reflection • Participate in onsite seatwork (3) on Diophantine equations • Attend onsite class for the discussion of Diophantine equations		1.5		1.5

	Week 14 Nov 18 -23	• Gospel Reading, Prayer and Reflection • Study uploaded materials on Module 6: Congruences Properties of congruences • Watch uploaded supplementary videos about congruences	Schoolbook MS Teams		1.5	1.5
		• Gospel Reading, Prayer and Reflection Take online Enabling Assessment (6): on Diophantine equations		1.5		1.5
	Week 15 Nov 25 -30	• Gospel Reading, Prayer and Reflection • Study uploaded materials on Module 6: • Watch uploaded supplementary videos properties of congruences	Schoolbook MS Teams		1.5	1.5
		• Gospel Reading, Prayer and Reflection • Attend onsite class for the discussion properties of congruences		1.5		1.5
	Week 16 Dec 2-7	• Gospel Reading, Prayer and Reflection • Study uploaded materials on Module 6: Linear Congruences • Watch uploaded supplementary videos about linear congruences	Schoolbook MS Teams		1.5	1.5
		• Gospel Reading, Prayer and Reflection Take onsite Enabling Assessment (7): Onsite quiz on linear congruences		1.5		1.5
Module 7- Chinese Remainder Theorem TLO12 TLO13 CLO3 CLO5 CLO6	Week 17 Dec 9-11	• Gospel Reading, Prayer and Reflection • Module 7 consists of Chinese Reminder Theorem • Study the uploaded materials for Module7 • Watch uploaded supplementary videos about Chinese Remainder Theorem Take online Enabling Assessment (8): Online quiz on Chinese Remainder Theorem	Schoolbook MS Teams		1.5	1.5
		• Gospel Reading, Prayer and Reflection • Attend onsite class for discussion and participate in onsite seatwork (4) on Chinese Remainder Theorem		1.5		1.5
	Week 17-18 Dec 12-14 Dec 16-18	Review topics under Modules 4 – 6			1.5	1.5
		Summative Assessment 2: Final Exam Onsite Paper and Pen Exam on Topics Modules 4 – 6		1.5		1.5

TOTAL	**10.5**	**13.5**	**27.0**

GRADING SYSTEM:

Midterm	%	Final	%
Enabling Assessments (Activities, Quizzes)	55%	Enabling Assessments (Activities, Quizzes, Output Presentations)	55%
Summative Assessment (Midterm Examination)	30%	Summative Assessment (Final Examination)	30%
Class Participation (classroom recitation, game activities)	15%	Class Participation (classroom recitation, game activities)	15%
	100%		100%

FINAL GRADE = $\frac{\text{Midterm Grade + Final Grade}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. Office365 Activation. Please ensure that your Office365 account is working. Your Office365 account is needed to access both Schoolbook and MS Teams where your asynchronous classes will be held.

2. Enrollment in an E-Class. You will automatically be enrolled in your e-class which is based on your enrollment data.

3. Traditional Blended Modality. This course adopts the traditional blended model. This means that there will be a mix of face-to-face, and asynchronous activities or online classes. This is divided into fifty percent face-to-face and fifty percent online.

4. Online Asynchronous Sessions.
    a. Schoolbook (SB). Schoolbook shall be the only platform for asynchronous sessions.
    b. Modules. Modules are self-paced learning resources for asynchronous sessions. These can be accessed in Schoolbook.
    c. References. Each page section may contain uploaded references. These learning resources may be downloaded.
    d. Asynchronous Activities. You are expected to read the modules as soon as they are uploaded. The learning content of the modules complement the online synchronous and face-to-face sessions.
    e. Asynchronous Engagement. Your activities in the course can be tracked by your professor. This includes the time you spend in reading the lessons and answering the assessments.
    f. Schoolbook Forum. All general concerns about the lessons and assessments in asynchronous sessions must be posted in the Schoolbook Forum. Response shall be made by your teacher within 48 hours.
    g. Schoolbook Messaging. This shall be the mode of communication for private and/or confidential communications. Response shall be made by your teacher within 48 hours upon receipt of the same unless it falls on weekends or holidays, which shall be handled promptly the following working day.

5. Face-to-face Sessions.
    a. Face-to-face engagement. Your engagement in face-to-face classes is graded based on your class participation.

b. Gospel Reading and Prayer. Each face-to-face session shall start with a Gospel reading and prayer. Your teacher may assign you, in advance, to do this.

c. Face-to-face Meeting Schedule. The meeting schedule shall follow the time indicated in your official registration.

d. Attendance. Attendance in face-to-face meetings is required.

e. Tardiness. A student who came one (1) minute to 25 minutes after the start of the face-to-face meeting is considered late. Three (3) tardy attendances are equivalent to one (1) absence.

f. Absence. Students who came more than 25 minutes after the start of official class schedule are considered absent. Incurring accumulated absences of more than 20% of total hours of face-to-face classes for the entire semester means a grade of 0.0 as final grade. This means that for a 3-unit subject, the maximum allowable absences in face to face classes, inclusive of tardiness, is five (5) hours.

g. Excuse from face-to-face classes. Students are excused in the face-to-face classes based on the provisions in the latest version of the Student Handbook.

h. Dress code. Attire should follow the policy on dress code as stipulated in the latest version of the Student Handbook.

6. Assessment and Grading System.

a. Enabling assessments. These will comprise most of your graded assessments. These are designed to achieve topic learning outcomes, which leads towards achieving the course learning outcomes. Only one enabling assessment shall be allowed during the week.

i. Online Enabling Assessment

Online enabling assessment shall be uploaded on SB and shall be accessible to students for 48 hours, while output-based enabling assessments shall be open for at least 6 days.

ii. Onsite Enabling Assessment

At least one enabling assessment per term shall be given onsite.

b. Summative assessments.

i. There shall be two summative assessments (midterm and final exams) for the entire semester. These are designed to achieve the course learning outcomes.

ii. Summative assessment shall be given through face-to-face.

iii. Output-based summative assessment shall be given to students at least fifteen days prior to schedule of Summative Exam Week.

c. Number of Attempts. All online enabling assessments shall have a minimum of 2 and a maximum of 3 attempts. Online summative assessments shall have two attempts.

d. Missed Assessments. Only students, (1) with valid reason as stated in the Student Handbook and CBL, and (2) who have given their proof of excuse on or before the next F2F session, shall be given a lifeline on the enabling and summative assessments.

i. An approved make-up enabling assessment shall be given onsite.

ii. The student must secure a special exam permit prior to taking onsite summative assessment.

e. Rubric. All online non-quiz or non-discrete types of assessment (essay, drop box, output-based, etc.) shall have a rubric or criteria for rating the students' tasks. A student may refuse to answer these types of assessment in the absence of a rubric or criteria for grading, and the assessment shall be deemed invalid and shall not be part of the student's grades.

f. Grading. All online assessments should be checked and graded by the teacher before the submission of midterm and final grades.

g. Gradebook. Students can see the breakdown of grades in their Assessment tab.

7. Synchronous Week.

a. Synchronous Week. It refers to the week following the midterm examination of each semester during which all classes are conducted live online, requiring students to attend at specified times according to their course schedule.
b. Schedule. The synchronous week falls on Week 10, October 21 – 26, 2024.
c. Platform. MS Teams will serve as the official platform for synchronous classes.

8. Independent Learning Days (ILDs).
    a. Goals. The goals of Independent Learning Days are to promote self-discipline and responsibility, foster critical thinking and problem-solving, encourage creativity and innovation and support personalized learning.
    b. Purposes. ILDs can be used for various purposes, such as: Individualized Learning, Research and Inquiry, Creative Expression and Skill Development.
    c. Schedule. ILDs are set on Week 11, October 28 – November 2, 2024.
    d. Modules and Assessments. Modules may be uploaded, and assessments may be administered online during ILDs.

REFERENCES:

Online References

Call number or e-provider	Reference Material
Open Access Resources	Elsevier B.V. 2019. Journal of number theory. [accessed 2024 August 5]. https://www.journals.elsevier.com/journal-of-number-theory
Open Access Resources	Hefferon J. 2002. Number theory. [accessed 2024 August 9]. https://jheffero.w3.uvm.edu/proofs/ibl-compact.pdf
Open Access Resources	Hardy, G. H 2007. An introduction to the theory of numbers. [accessed 2024 July 10]. https://projecteuclid.org/euclid.bams/1183493592
EBSCOhost	Deng, X., Jiang, W. 2015. Number theory based game-theoretic framework in adversarial decision making under a fuzzy environment. In International Journal of Approximate Reasoning. 106:194-213.

On-Site References

Call number or e-provider	Reference Material
QA 241 .W325 2014	Watkins, J. 2014. Number theory: a historical approach. Princeton (NJ): Princeton University Press.
QA 241 .Es74 2017	Borres, MC. 2017. Essentials of number theory. New York (NY): Arcler Press LLC.
QA 241 .C817 2015	Corry, L. 2015. A brief history of numbers. New York (NY): Ohio University Press.

QA 241 .St49 2015	Stewart, I. 2015. Professor Stewart's incredible numbers. New York (NY): Basic Books

Prepared: **Mathematics and Statistics Department**
1st Semester 2024-2025

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